

Multinomial and Ordinal Responses

A response with two categories has one logit. A response with c categories has $c - 1$ of them, and the modelling question is which $c - 1$ comparisons to make. If the categories are names — a brand, a party, a mode of travel — the natural comparisons are with a fixed baseline, and the result is the *baseline-category logit model*, a generalized linear model whose response is a vector of counts. If the categories are ordered, that choice wastes information: a model ignoring the order needs $(c - 1)p$ coefficients to say what a single slope may say. Ordinal models get back down to p by cutting the scale and giving every cut the same coefficients. Where the cut is made is the whole story. Cut *at* a level and you get the cumulative, or proportional odds, model; cut *conditionally on having got that far* and you get the sequential, or continuation-ratio, model. These answer different questions, and the data can tell them apart.

What you need. [Chapter 34](#) (exponential families, the likelihood equations, Fisher scoring, the deviance and the three test statistics) and [Chapter 35](#) (logistic regression, the alternative links, separation, goodness of fit). From earlier parts: [Chapter 11](#) for the likelihood ratio test Theorem 11.4.1 and the three flavours of test statistic Proposition 11.4.2, [Chapter 21](#) for weighted least squares and the sandwich estimator (Theorem 21.4.2), and [Chapter 29](#) for the information criteria (Definition 29.2.2). Nothing here needs normality of anything.

36.-1.1 Roadmap

- [Baseline-category logits](#)
- [Cumulative logit models](#)
- [Proportional odds and its checks](#)
- [Sequential models](#)
- [Summary and notes](#)